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CO₂-Modified Surfactant for CO₂ Sequestration in Carbonate Reservoir

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Abstract

A CO₂-to-chemical method was proposed for CO₂ sequestration by injection of the CO₂-based chemical solution into carbonate reservoirs. The CO₂ modification of a non-ionic alkyl polyglycoside (APG) surfactant was conducted using a glass reaction vessel in a reaction equipment under gentle conditions at a temperature of 50°C and a CO₂ pressure of 2 bar. In the presence of 1,8-diazabicyclo [5.4.0] undec-7-ene (DBU) and using dimethyl sulfoxide (DMSO) as a solvent, CO₂ reacted with the hydroxyl (-OH) groups of the APG structure, especially the -OH group attached on the C6 position of the glucose unit, to transfer the non-ionic -OH group to the anionic -COO- functional group. By this approach, CO₂ was successfully incorporated into the APG molecule to yield a CO₂-modified APG (CMAPG) product to effectively reduce CO₂ gas emission. The CO₂ consumption capacity was estimated to around 0.13 ton of CO₂ per ton of APG raw product, which is higher than most of the recently developed CO₂ absorbers. This CO₂-involved reaction was not reversible by external stimuli and the CMAPG surfactant was thermally stable at 50°C, suggesting that CO₂ will not be released during sequestration. The new surfactant also demonstrates an enhanced properties in terms of interfacial tension (IFT) reduction and adsorption on carbonate. The IFT between crude oil and CMAPG solution decreased by approximately ten-fold compared with that obtained from the APG solution. Both static and dynamic adsorption of CMAPG onto carbonate are higher than the adsorption of the APG precursor, indicating a significant potential in CO₂ sequestration by injecting the CMAPG solution into depleted carbonate reservoirs.

Introduction

The escalating threat of global climate change, primarily driven by rising levels of atmospheric carbon dioxide (CO₂), has necessitated the deployment of technologies aimed at capturing and securely storing CO₂ emissions. Among these, carbon capture and storage (CCS) is recognized as a pivotal mitigation strategy, particularly for decarbonizing hard-to-abate sectors such as power generation, cement production, and steel manufacturing. Central to the success of CCS is the identification and utilization of suitable geological formations for long-term CO₂ sequestration (Wang et al., 2022). In this regard, depleted carbonate reservoirs

—which have already served as sites for hydrocarbon extraction—are emerging as promising candidates (Pan and Mohanty, 2024).

Carbonate reservoirs, predominantly composed of calcium-rich minerals such as calcite and dolomite, offer several advantages for CO₂ sequestration. These include existing infrastructure, known reservoir characteristics, and proven caprock integrity, all of which can significantly reduce operational costs and uncertainty. Additionally, many of these reservoirs are already well-studied, with extensive geological, petrophysical, and fluid-flow data accumulated during decades of production. Their high porosity and residual pore space make them suitable for storing significant volumes of CO₂, and in some cases, the process can be coupled with enhanced oil recovery (EOR) to improve project economics (Lv et al., 2022; Tan et al., 2022).

Despite their advantages, the implementation of CO₂ sequestration in carbonate formations is accompanied by a set of complex technical, geochemical, and operational challenges, which must be addressed to ensure safe and effective storage. One of the primary difficulties arises from the intrinsic heterogeneity of carbonate reservoirs. Unlike siliciclastic formations, carbonate rocks often exhibit variability in porosity and permeability due to factors such as diagenesis, fracturing, and dissolution features. This spatial heterogeneity complicates the prediction of CO₂ plume migration, storage efficiency, and potential leakage pathways (Orivri et al., 2025; Sohal et al., 2021; Ye et al., 2023; Zhou et al., 2024).

Another major concern is geochemical reactivity. When CO₂ is injected into a carbonate formation, it dissolves into formation water, forming carbonic acid. This acidic fluid can react with carbonate minerals, leading to dissolution of the rock matrix, which may increase porosity but reduce mechanical integrity. Conversely, it can also lead to mineral precipitation in certain zones, clogging pore spaces and reducing injectivity. These coupled geochemical and mechanical processes introduce significant uncertainties in reservoir behavior and require robust reactive transport modeling to anticipate long-term impacts (Orivri et al., 2025; Sohal et al., 2021; Ye et al., 2023; Zhou et al., 2024).

Furthermore, the presence of natural fractures and faults in carbonate reservoirs can pose risks to CO₂ containment. Fractured systems can act as high-permeability pathways, accelerating CO₂ migration toward caprocks or fault zones and increasing the potential for leakage. Assessing the sealing capacity of these structures and their response to changes in pressure and geochemistry is critical for risk management (Orivri et al., 2025; Sohal et al., 2021; Ye et al., 2023; Zhou et al., 2024).

Monitoring and verification in carbonate reservoirs also present unique difficulties. The interpretation of geophysical signals (e.g., seismic, electromagnetic) can be challenging due to the complex lithology and fluid interactions in these systems. As such, deploying effective monitoring technologies that can detect and track CO₂ movement with high resolution remains an ongoing area of research (Orivri et al., 2025; Sohal et al., 2021; Ye et al., 2023; Zhou et al., 2024).

This work explores a new method related to CO₂ sequestration in depleted carbonate reservoirs by incorporating CO₂ into alkyl polyglucoside (APG) surfactant molecules. After reacting with CO₂, the CO₂ molecules attach to the hydrophilic glucose units by covalent bonds and convert the non-ionic surfactant into a partially anionic surfactant, thus facilitating the interaction between the CO₂-modified APG (CMAPG) surfactant with carbonate surface via electrostatic attraction. In this approach, as CO₂ is not directly sequestered in the form of gas, the potential of leakage by CO₂ migration is significantly reduced because the CO₂ is "fixed" in a chemical. This study aims to contribute to the optimization of carbonate reservoirs as viable long-term and permanent CO₂ storage sites and their integration into global decarbonization strategies.

Experimental

Chemicals and brines

The APG surfactant was obtained from Shanghai Fine Chemical Co Ltd (Shanghai, China). The chemical data are listed in Table 1. The catalyst 1,8-diazabicyclo [5.4.0] undec-7-ene (DBU) and the solvent dimethyl sulfoxide (DMSO) were purchased from Innochem Co Ltd (Beijing, China). The gas cylinder filled with high-purity CO₂ gas was supplied by AirProducts (Beijing, China). Deionized (DI) water was produced from a Millipore Milli-Q system and used to prepare two brines with a total dissolving solid (TDS) of 213,734 ppm to mimic the formation water (FW), and a TDS of 57,670 ppm to mimic the injected water (IW), respectively. The compositions of both synthetic brines are listed in Table 2.

Table 1—Information on the APG1214 sample.

Trade name	Number of the carbon on alkyl chain	Number of the glucose units	Effective content	Hydrophilic-lipophilic balance
APG1214	12-14	1-2	50%	11.3

Table 2—Compositions of FW and IW.

Ions	Ion concentration of the FW, ppm	Ion concentration of the IW, ppm
Na ⁺	59,491	18,300
Ca ²⁺	19,040	650
Mg ²⁺	2,439	2,110
SO ₄ ²⁻	350	4,290
Cl ⁻	132,060	32,200
HCO ₃ ⁻	354	120
TDS	213,734	57,670

Synthesis apparatus

The CO₂ modification was conducted using a high-temperature and high-pressure (HTHP) reactor (YZTR-300Q) manufactured by Shanghai Yan Zheng Experimental Instruments Co Ltd). A 200 mL reaction vessel made by glass (maximum pressure 20 bar) was equipped for visual observation of the reaction.

Stability test

The stability of the CO₂-modified surfactant was evaluated by the interfacial tension (IFT) between crude oil and aqueous phase of the surfactant solution after aging for various period at 50°C. A dead light crude oil from a carbonate reservoir was used. IFT was determined using a spinning drop tensiometer (SVT20, Dataphysics) at the aging temperature and a rotation speed of 5,000 min⁻¹.

Characterization

The CO₂ modification was confirmed by comparing the FTIR spectra of the reaction mixture before and after exposure to CO₂. UV-vis absorption spectra were obtained by a UV-vis spectrophotometer (Shimadzu UV-2600). According to the Beer-Lambert Law, absorbance of the sample at a given wavelength can be expressed by Eq. 1

$$A = \varepsilon bc \quad (1)$$

where A represents the absorbance at a specific wavelength, ε is molar extinction/absorption coefficient at the wavelength, b is the path length of the cuvette (1 cm), and c is the surfactant concentration. By plotting the absorbance versus concentration, a linear relationship was obtained as the calibration curve. To avoid oversaturation of the acquired UV-vis spectra, all absorbance values at the specific wavelength are smaller than one.

Static adsorption

The static adsorption of surfactants on carbonate surface was determined by Eq. 2. The carbonate fine powder was prepared by sieving the crushed and grinded core plug with a mesh size of 60-80. The APG and CMAPG surfactants were dissolved in the IW to prepare a series of solutions at concentrations ranging from 500 to 5,000 ppm. The carbonate powder was immersed in each surfactant solution at a mass ratio of 1:5 and the mixtures were placed in an oven at 50°C for 24 hours. After centrifuging the supernatant for 20 mins at 5,000 rpm, the upper solution was taken for the UV-vis measurement. The surfactant concentration at equilibrium of adsorption was calculated from the peak absorbance using Eq. 1.

$$\Gamma_s = \frac{(C_o - C_e) \times V}{M_{rock}} \quad (2)$$

where Γ_s represents the static adsorption in mg/g rock, C_o and C_e are the surfactant concentrations in g/L before adsorption and at equilibrium, respectively. V is the solution volume in L, and M_{rock} is the mass of the carbonate powder in g.

Dynamic adsorption

Core-flooding tests were used to investigate the dynamic adsorption of surfactants. A schematic diagram of the experimental setup is shown in Fig. 1. The experimental procedures in details are listed below:

- The carbonate cores were dried at 95°C for overnight.
- The weight, size, and gas permeability of the dried cores were measured.
- Immersed the core plugs in filtered FW for overnight under vacuum.
- A core plug saturated with the FW was weighed and loaded into the core-flooding equipment
- Determined the brine permeability using Darcy's law.
- Injected the surfactant solution at 2,000 ppm in the IW for 5 PV at 50°C using a flow rate of 0.5 mL/min.
- Injected the IW at same conditions to mimic subsequent water flooding.
- Collected the produced fluids after a constant time interval for determination of surfactant concentration.

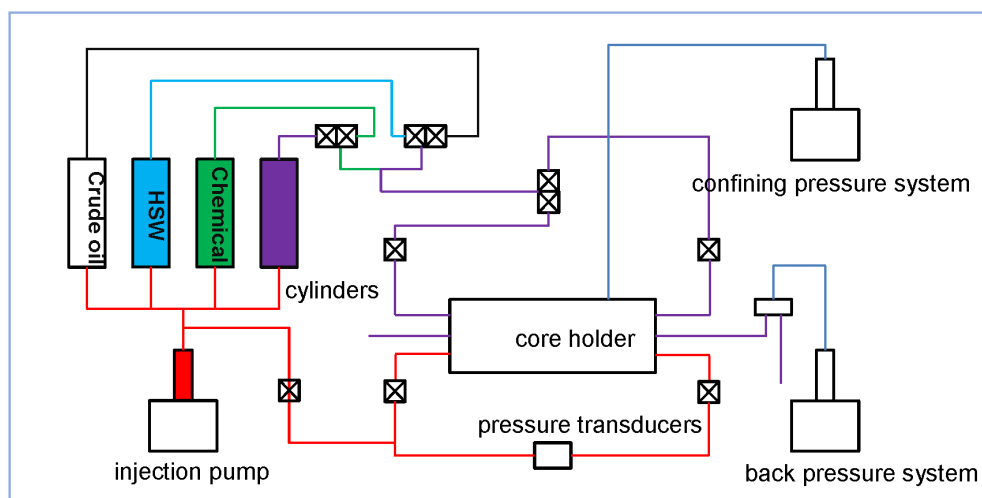


Figure 1—Schematic diagram of core-flooding experimental setup.

Table 3 lists the parameters of two carbonate core plugs. The surfactant concentrations after adsorption were determined by UV-vis spectra. The dynamic adsorption can be obtained using Eq. 3.

$$\Gamma_d = \frac{C_o \times V_o - \sum_{i=1}^N C_i \times V_i}{M_{rock}} \quad (3)$$

Table 3—Core parameters.

Core #	Injected surfactant	Diameter (cm)	Length (cm)	Pore volume (mL)	Gas permeability (mD)	Brine permeability (mD)
1	APG	3.844	7.048	9.56	55.2	26.7
2	CMAPG	3.813	7.050	10.25	53.9	26.4

where Γ_d represents the dynamic adsorption in mg/g rock, C_o is the surfactant concentration in g/L before adsorption, V_o is the total injection PV in L. C_i and V_i represent the surfactant concentration in each produced fluid in g/L, and volume of each collection in L, respectively. N is the number of fluid collection, M_{core} is the mass of the carbonate core in g.

Evaluation of CO₂ consumption

Certain amounts of APG were dissolved into DMSO in 20-mL vials under stirring to prepare a series of solutions with various concentrations. Based on the information of APG product listed in Table 1, the average molecular weight of APG was estimated to be 450 g/mol. Six-fold excess in mole of DBU was injected via a syringe into the vials with different APG concentrations under stirring. After DBU was fully dissolved, the vial was sealed with a rubber septum and loaded into a water bath at 50°C. Two needles were inserted into the vial with one immersed into the solution and close to the bottom of the vial for CO₂ bubbling, and the other one placed above the solution for vent. The entire experimental setup including solution, vial, septum, needle, stir bar and water bath was weighed as the initial mass. CO₂ was then bubbled into the solution at a constant rate under stirring to maintain a stable pressure of 2.5 bar. After a white precipitate was observed, CO₂ bubbling was removed and the experimental setup was weighed as the final mass. The increase in mass corresponds to the amount of CO₂ sequestered in the APG solution. Considering the CO₂ dissolved in DMSO, the data obtained from pure DMSO solvent was used as a reference. Each experiment was repeated for three time to obtain the standard deviation.

Results and Discussion

CO₂ modification and synthesis of CMAPG

The non-ionic APG surfactants exhibit excellent eco-friendly properties because they degrade into glucose and fatty alcohols, both of which are naturally occurring and environmentally benign. Furthermore, they show very good thermal stability and compatibility in high-salinity water, robust surface and interfacial activity, and synergistic effects when combined with other oilfield chemicals. Therefore, APG surfactants have gained much more attention as a promising agent for enhanced oil recovery (EOR) applications in the past decades (Chen et al., 2022; 2023; 2024a; 2024b; 2025; Iglaier et al., 2009; 2010; Wei et al., 2018; 2020).

Fig. 2a shows a reaction scheme on synthesis of CMAPG by reacting CO₂ with APG surfactant in DMSO in the presence of DBU. After the reaction, CO₂ transferred the hydroxyl group of APG to a -COO- group and converted the non-ionic APG into a partially anionic surfactant. The -OH group at the C6 position is mostly reactive compared with the other two connected with the C2 and C3 positions. The substitution occurs at the C2 or C3 positions is difficult due to the steric hinderance as both -OH groups are so close to

the bulky glucosyl structure of APG (Miljković, 2010). Thus, we assume that one mole of APG reacts with 1-2 moles of CO₂. The amounts of the reactants were designed based on this principle.

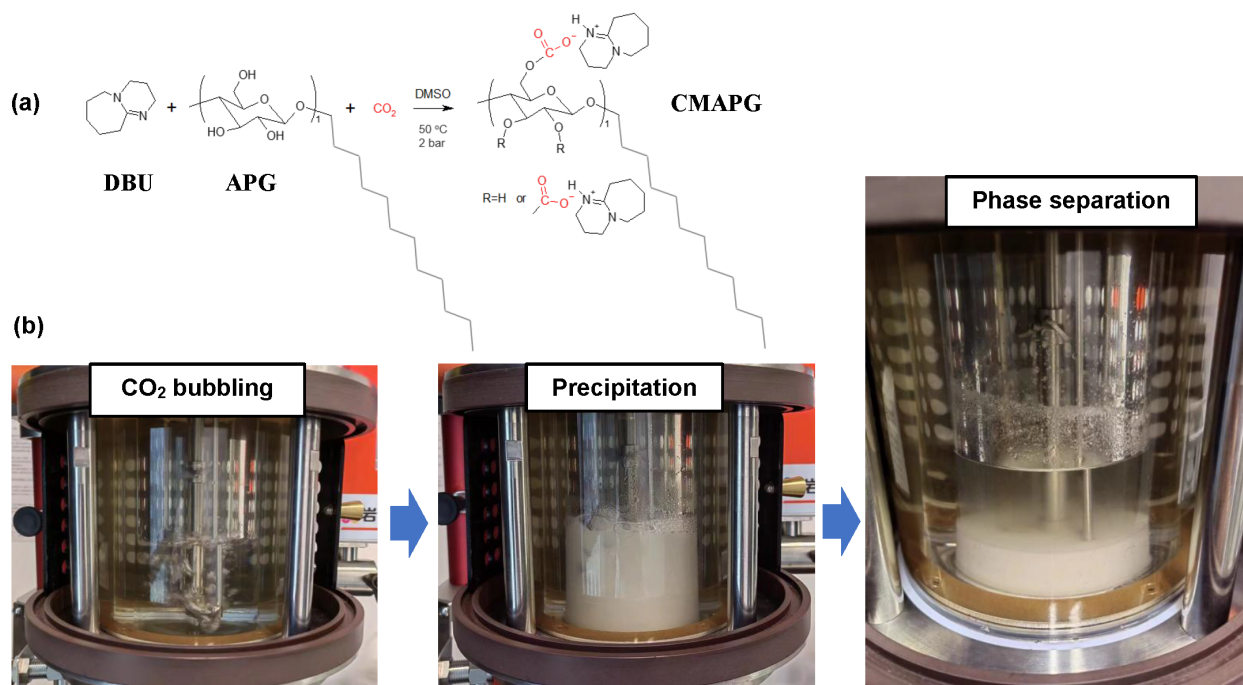


Figure 2—Reaction scheme (a) and preparation of CMAPG in the HTHP reaction system (b).

Before CO₂ modification, 5 g of APG was dissolved in 100 mL DMSO, then DBU in three-fold molar excess was added in the solution followed by continuous stirring until a homogeneous and clear solution was obtained. The solution was transferred into the reaction vessel at a temperature of 50-60°C. CO₂ was introduced into the sealed reaction vessel by bubbling in the solution to maintain a CO₂ pressure of 2-3 bar. After stirring for several hours, white precipitation was observed due to formation of glucose acid which is not soluble in DMSO. After the reaction, the suspended solution was centrifuged for 20 min at 3,000 rpm. The supernatant was discarded, and the white waxy solids were collected as the final CMAPG products. Fig. 2b indicates all above observations during the synthesis.

The CO₂ modification of APG was confirmed by FTIR characterization. Fig. 3 shows the FTIR spectra of the APG/DBU/DMSO mixture before and after the CO₂ reaction. The spectrum in black indicates a single and sharp peak at 1630 cm⁻¹, corresponding to the stretching band of C=N-C group from DBU. After the mixture exposed to a CO₂ environment by CO₂ bubbling, the peak was broadened due to formation of new ester group which is characterized by the adsorption band at 1720 cm⁻¹. The CO₂ directly dissolved in DMSO solvent also shows a new and broad adsorption band at around 2600 cm⁻¹. Both new peaks from the FTIR spectrum in green are indicated by the red arrows. Fig. 3 also shows that the -OH groups characterized by the peak at 3400 cm⁻¹ still exist after the reaction, suggesting that some of the hydroxyl groups of the APG surfactant remained unmodified due to the incomplete substitution of these groups attached on the C2 or C3 positions, which agrees with the reaction scheme shown in Fig. 2a. Generally, the peak at 1720 cm⁻¹ provides a clear proof on successful modification of APG by CO₂.

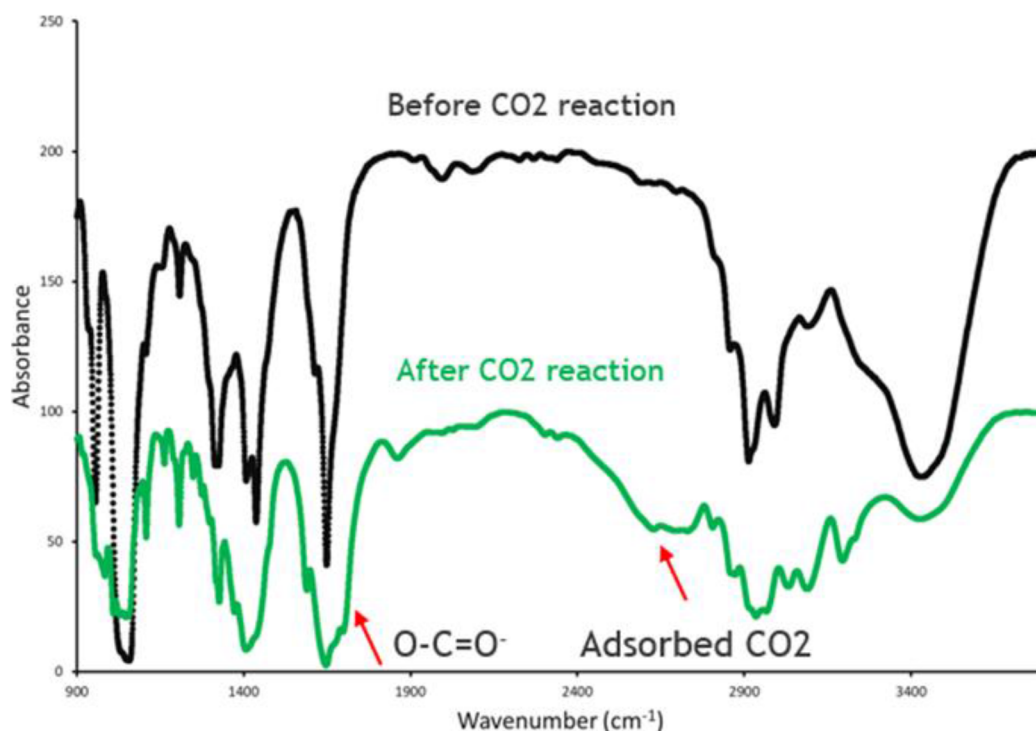


Figure 3—FTIR spectra of the reaction mixture before and after CO₂ exposure.

The UV-vis spectra shown in Fig. 4 also indicate significant differences between APG and CMAPG surfactants. First, a blue shift is observed on the peak absorbance after the CO₂ reaction, as APG surfactant exhibits an absorbance peak at the wavelength of 221 nm while CMAPG demonstrates a peak absorbance at 213 nm. Second, the extinction/absorption coefficient at the wavelength when absorbance peaks almost doubles after the APG surfactant was modified by CO₂. At the surfactant concentration of 1,250 ppm, the absorbance of APG is 0.461 while the value increases to 0.785 for CMAPG. This result suggests that the CMAPG product has a stronger capacity in adsorbing UV light, probably due to the incorporation of the conjugated structures from DBU as indicated by the reaction scheme.

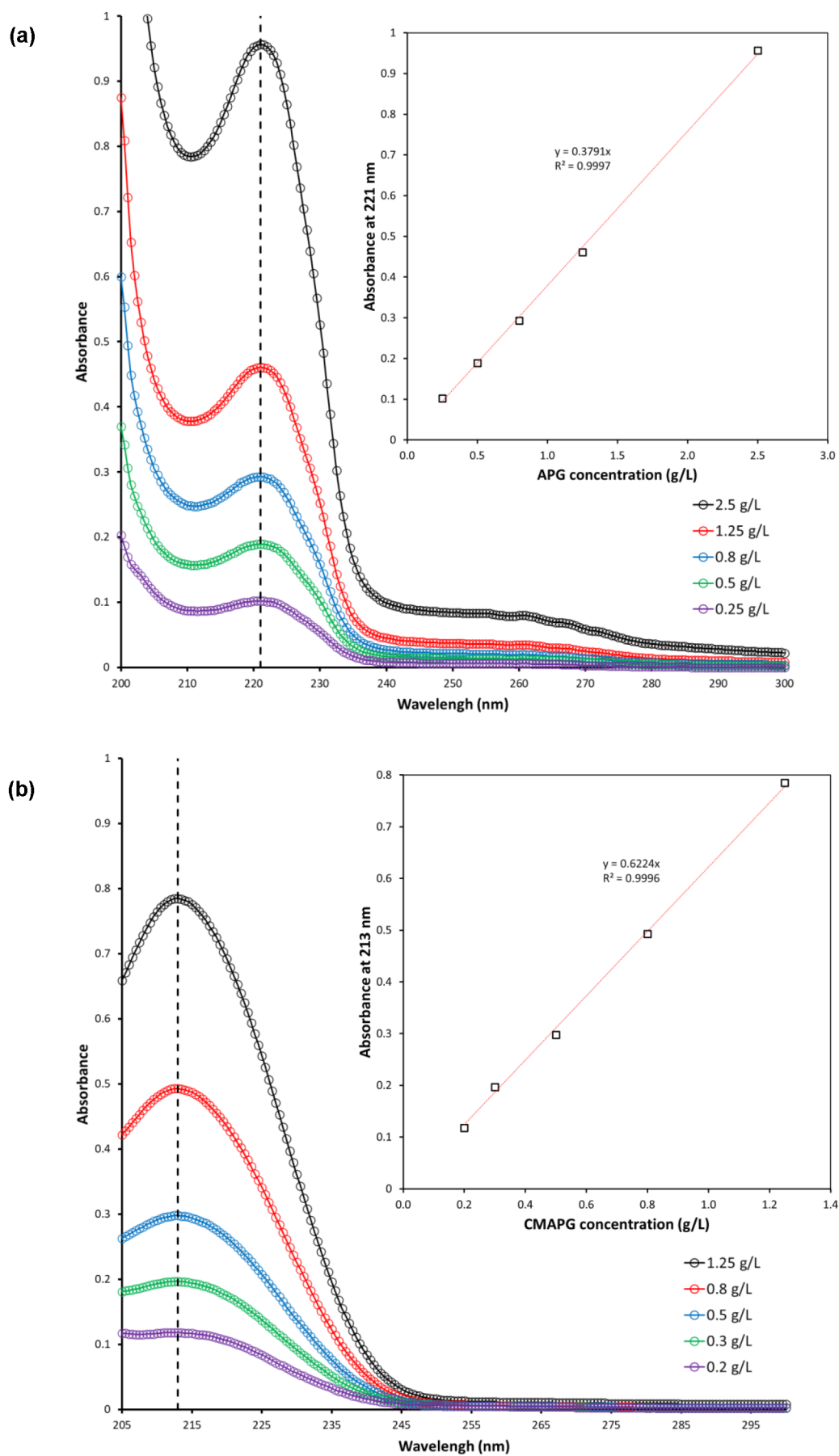


Figure 4—UV-vis spectra of APG (a) and CMAPG (b) in the IW at different concentrations, inset: baselines of absorbance at 221 nm for APG and 213 nm for CMAPG, respectively.

Stability of CMAPG

It should be noted that the CO₂ reaction and modification introduced in current study is totally different with the CO₂-triggered switchable systems which demonstrated a clearly defined reversible route by external stimuli such as altering temperature or passing through an inert gas. For example, the CO₂-viscosifying solution forms a gel by bubbling with CO₂ and maintains an extremely high viscosity in a CO₂ environment while the system switches back to a very low viscosity in a nitrogen atmosphere (Su et al., 2013; Yang et al., 2019). Comparatively, the CMAPG product was unaffected by raising the ambient temperature to 100°C or passing through nitrogen gas for half an hour, suggesting that the CO₂ modification is not a reversible reaction (Chen et al., 2025).

However, it is still necessary to evaluate the stability of CMAPG to guarantee an effective CO₂ sequestration under reservoir conditions. This was conducted by determination of IFT between crude oil and the surfactant solutions after aging at 50°C. Hou et al (2024) applied the IFT data collected during various aging times from six surfactants to analyze the thermal stability under "harsh" conditions. The IFT results obtained with APG and CMAPG solutions at 2,000 ppm as a function of aging time are shown in Fig. 5. First, the IFT of APG surfactant was significantly reduced by ten-fold from around 0.5 mN/m to 0.05 mN/m after the CO₂ modification, indicating a significant potential for EOR practice. Second, the IFT obtained from CMAPG remain constant up to 90 days of aging, suggesting that no changes in the chemical structure by decomposition or degradation during the aging test at 50°C. Particularly, the ester-like -COO- bonds from CMAPG structure are considered not very stable, as they tend to degrade at higher temperature to convert CMAPG to the initial APG structure to release CO₂. But this conversion did not occur because the IFT of CMAPG remained unchanged over the entire aging period. Visual observation was also conducted with a CMAPG solution in the IW, the solution was clear after aging for three months, indicating that CMAPG was thermally stable and can be used as a chemical agent for EOR and CO₂ sequestration at 50°C.

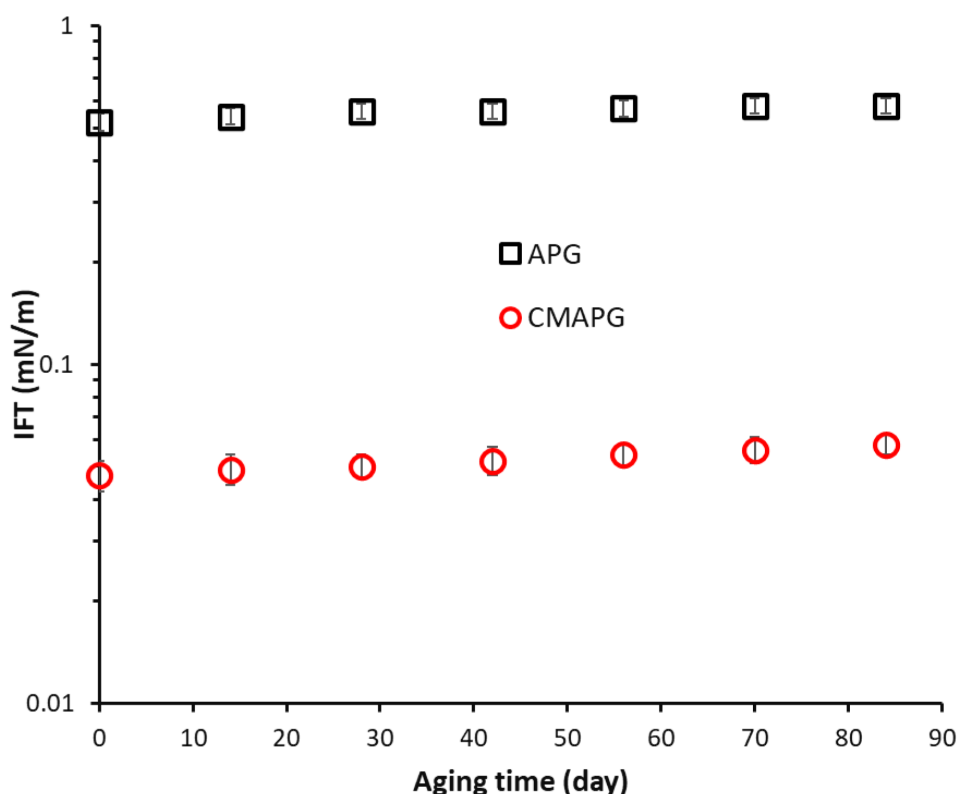


Figure 5—IFT as a function of aging time of APG and CMAPG at a surfactant concentration of 2,000 ppm.

Static adsorption

Understanding surfactant retention within the target reservoir formation is critical for optimal chemical selection and formulation design. During chemical flooding operations, a portion of the surfactant may be retained in the formation due to various mechanisms, including adsorption onto the rock surface, precipitation, phase trapping, and partitioning into the oil phase (Kalam et al., 2021). High levels of surfactant retention necessitate the injection of larger chemical slugs, which increases operational costs and reduces economic feasibility. Surfactant precipitation and phase trapping are typically the result of chemical incompatibility under harsh reservoir conditions. Surfactant adsorption primarily arises from interactions between the rock surface and the hydrophilic head groups of the surfactant molecules, mediated by electrostatic forces and hydrogen bonding (Ahmadi and Shadizadeh, 2015). Table 4 lists the static adsorption data for APG and CMAPG at various concentrations using carbonate rock powder. At a concentration of 2,000 ppm, the static adsorption values for APG and C-APG were found to be 0.93 and 1.08 mg/g of rock, respectively. The slightly higher adsorption of CMAPG is attributed to its partial anionic character, acquired through CO₂ modification, which facilitates electrostatic attraction with the positively charged carbonate surface. In contrast, the non-ionic nature of unmodified APG limits such interactions.

Table 4—Static adsorption data of APG and CMAPG surfactants at various concentrations in the IW at 50°C.

Surfactant	Adsorption (mg/g carbonate rock)					
	500 ppm	1,000 ppm	2,000 ppm	3,000 ppm	4,000 ppm	5,000 ppm
APG	0.21	0.50	0.93	1.26	1.51	1.78
CMAPG	0.26	0.60	1.09	1.45	1.76	2.07

Dynamic adsorption

Fig. 6 illustrates the breakthrough curves representing the concentration ratio (C_i/C_o , where C_i is the surfactant concentration in the produced fluid and C_o is the initial concentration) as a function of injected PV. After an initial pre-flush using IW, a surfactant solution at 2,000 ppm was injected for 5 PV, followed by 5 PV of post-flush water. Surfactant concentrations in the effluent were measured from the onset of chemical injection. After approximately 3 PV, the concentration plateaued, indicating a steady-state condition. Based on the smoothed breakthrough curves in Fig. 6 and using Eq. 3, the dynamic adsorption values were calculated as 0.11 mg/g for APG and 0.13 mg/g for CMAPG. These values are considerably lower than those obtained under static conditions, suggesting that in a dynamic (flowing) environment, the adsorption driving forces are significantly weaker than the forces promoting desorption (Wei et al., 2018). This observation aligns with the findings by Ahmadi and Shadizadeh (2015), who reported that dynamic adsorption may be reduced to approximately one-tenth of the static value. Furthermore, the dynamic adsorption of surfactants can be further minimized when surfactant injection follows a pre-flush with polymer solution or when surfactants are co-injected with a surfactant-polymer mixture, due to competitive adsorption and flow-induced desorption mechanisms (Wang et al., 2015).

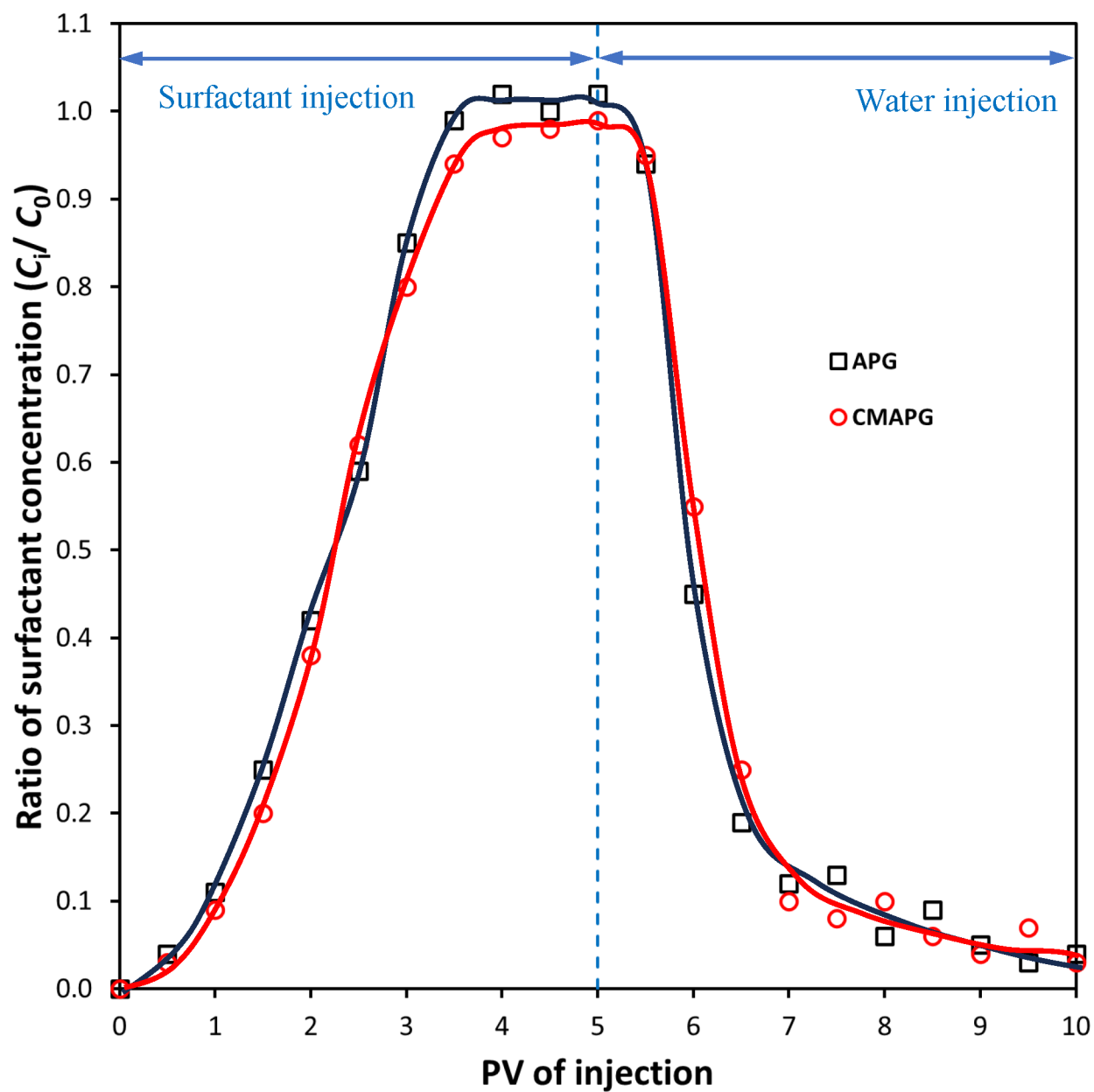


Figure 6—Surfactant concentration ratio versus injection PV of the core-flooding experiment.

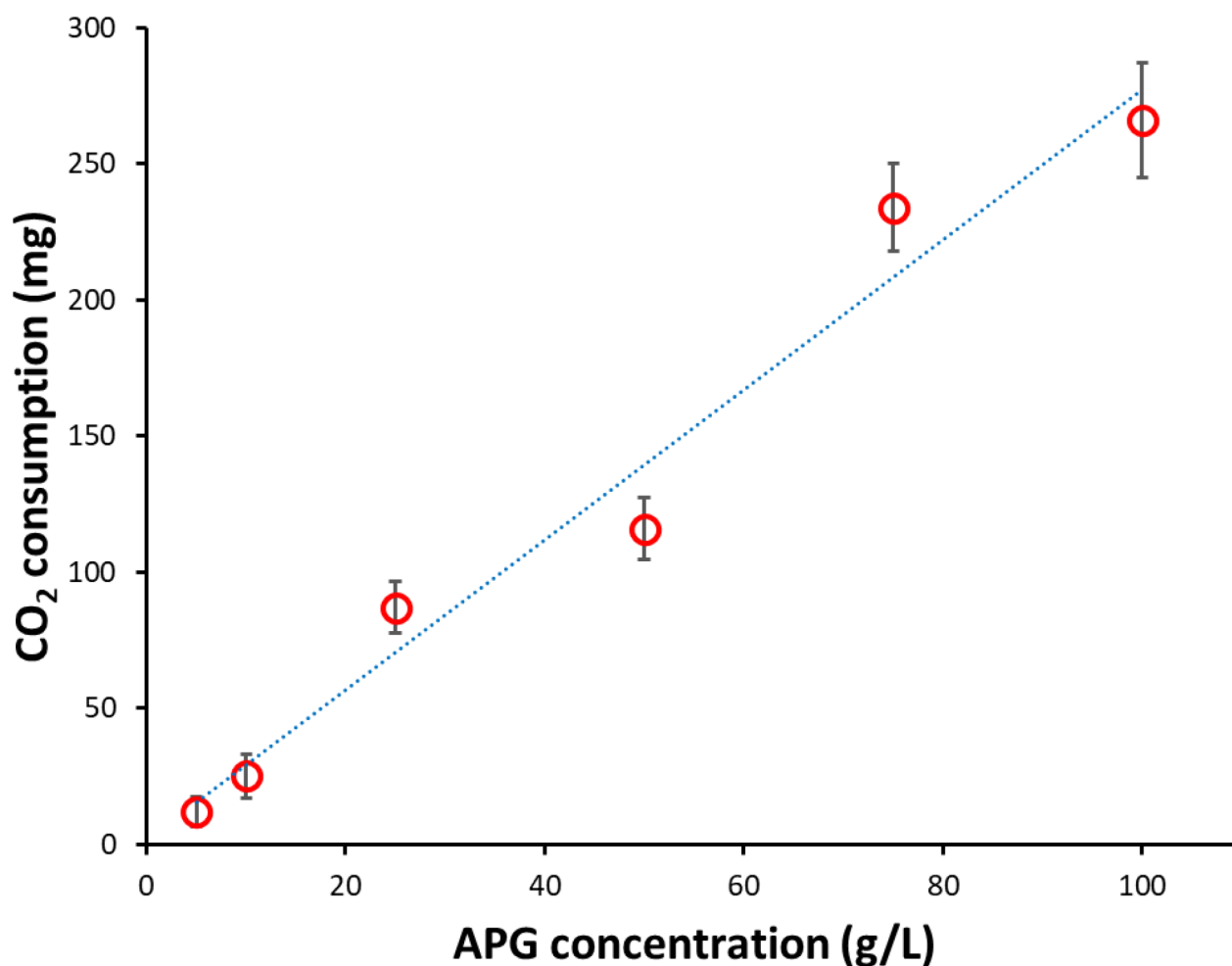


Figure 7—A plot and linear fit of CO₂ consumption amount as a function of APG concentration.

CO₂ reduction capacity evaluation

Six APG solutions in DMSO at concentrations of 5, 10, 25, 50, 75, and 100 g/L were prepared in the presence of six-fold molar excess of DBU to estimate the amounts of CO₂ uptake to evaluate the capacity of CO₂ sequestration. Fig. 15 indicates that the amount of consumed CO₂ (S) increases linearly with APG concentration (C_{APG}), the relationship is given in Eq 4.

$$S = (2.72 \pm 0.047)C_{\text{APG}} + (3.95 \pm 2.91) \quad (4)$$

Based on Eq 4, 1.0 mol of APG is able to sequester 1.39 ± 0.03 mol of CO₂, corresponding to that 0.13 ton of CO₂ can be eliminated by per ton of APG product. Because 1 mol of the initial APG sample contains 4–7 mol of -OH groups, the amount of CO₂ consumption is significantly lower than the theoretical estimation which predicts a 1:1 molar reaction ratio between CO₂ and the -OH group of APG. The reason could be attributed to different reactivity of the -OH groups at different positions, as described in introduction. Therefore, many unmodified -OH groups are present in the final C-APG product, which is in agreement with the FTIR characterization result. But nevertheless, such a sequestration capacity is still superior to most newly developed CO₂ absorbers as listed in Table 5.

Table 5—CO₂ capture capacity of traditional absorbents and newly developed materials.

Type	Name	CO ₂ capture capacity (mg/g)	Reference
Traditional adsorbents	aqueous ammonia	ca. 1000	Liu et al., 2009
	mono-ethanolamine	340	Blauwhoff et al., 1983
	di-ethanolamine	210	Blauwhoff et al., 1983
New materials	TMOF-1 (Metal-Organic Framework material)	65	Zhang et al., 2016
	NP30s (nanoparticle)	70	Werz et al., 2015
	b-PEI (polymer)	38	Kainz et al., 2015
	GP30s (microgel particle)	76	Yue et al., 2014
	PPN-6-SO ₃ Li (porous polymer network)	176	Lu et al., 2011
	Jeffamine (nanosilica)	78	Al-Azzawi et al., 2012
	LAHPs (amino acid salt encapsulated hydrogel)	73	Xu et al., 2021
	alkali/amine infused hydrogel	55-95	Xu and Wood, 2018
	Bile salts	40-45	Zhang et al., 2019
	APG	140	Current work

Conclusions

- CO₂ modification of a commercial APG product was conducted by reacting CO₂ with the APG and DBU in DMSO at a temperature of 50°C and a CO₂ pressure of 2 bar.
- FTIR and UV-vis spectroscopy confirmed the yield of CMAPG product.
- One mole of APG was able to consume 1.39±0.03 mol of CO₂, corresponding to a consumption capacity of 0.13 ton CO₂ per ton APG product.
- CMAPG exhibited thermally stable in the IW at 50°C.
- The oil/water IFT obtained from CMAPG was ten-fold lower than that of the APG surfactant.
- Both static and dynamic adsorption of CMAPG surfactant on carbonate were higher than that of the APG precursor.
- This study demonstrates a new approach for stable and permanent CO₂ sequestration in the carbonate reservoirs.

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